

Kairui (Gary) Ying

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EDUCATION

University of California, Berkeley

Berkeley, CA

B.A. in Computer Science and B.S. in Industrial Engineering | GPA: 3.94/4.00

Expected 2028

Relevant Coursework: Data Structures and Algorithms, Structure and Interpretation of Computer Programs, Linear Algebra, Probability, Foundations of Data Science, Full-Stack Web Development.

EXPERIENCE

Stanford University School of Medicine

Apr 2025 – Present

AI Engineer

Stanford, CA

- Shipped an agentic retrieval-augmented generation (RAG) platform to production for 2,400+ users, enabling natural-language querying over patient records, CVs, and Google Drive, secured by Stanford single sign-on (SSO) on Google Cloud (GCP) with audit logging on all personally identifiable information (PII) access.
- Built a permission-aware retrieval engine merging structured and semantic search into a single ranked context, running natural-language-to-SQL over a 7-table schema and pgvector HNSW vector search over 1,536-/3,072-dimensional embeddings in parallel behind a 5-way intent classifier.

PlatePromos

May 2024 – Present

CTO & Co-Founder

Encinitas, CA

- Built and deployed PlatePromos.com on Flask and AWS EC2, serving 500+ students and 35+ local business partners providing reduced-cost meals; recognized by the Mayor of Encinitas and the California State Senate (\$1,000 award).

Mirai Copilot

Jun 2025 – Jun 2026

Software Engineer

San Diego, CA

- Implemented principal component analysis (PCA) via eigendecomposition in NumPy, reducing Instagram post-embedding vectors from 1,536 to 128 dimensions for 40% faster nearest-neighbor retrieval in a hybrid RAG content-recommendation pipeline for creator accounts.

University of Michigan School of Public Health

Sep 2024 – Oct 2025

AI Research Assistant

Ann Arbor, MI

- Built a real-time kidney-discoloration classifier analyzing MRI scans via transfer learning in PyTorch, reaching 95% accuracy against radiologist-verified labels.

Gobattle.io – Shinobit

Aug 2023 – Sep 2025

Game Developer

Berkeley, CA

- Engineered a server-authoritative 2D browser massively-multiplayer online (MMO) game in vanilla JavaScript and HTML5 Canvas, holding 60 frames per second (FPS) via requestAnimationFrame for 50,000+ registered users.
- Built a Python WebSocket server for 1,000+ concurrent connections with grid-based broad-phase collision detection and delta-only state broadcasting, powering real-time player-versus-player (PvP), raids, and state synchronization across web, iOS, and Android.

PROJECTS

Autonomous Build Factory — factory-hub

Claude-based agents

- Designed and operate an autonomous build factory that ships one small finished project per day: spec-first builds, an adversarial multi-critic review gate, same-day live deploys, and a public audit trail at github.com/yinggarykairui/factory-hub.
- Factory-shipped projects include pixel-garden (procedural canvas garden, zero dependencies), trace-lens (TypeScript/React agent-trace replay tool), and orbit-doodle (physics drawing toy), each with a public repo and live demo under github.com/yinggarykairui.

MeetWise — AI Scheduling Platform

Node.js, Cloudflare Workers, Claude API

- Built a production scheduling app with server-side Google OAuth2 token exchange in a Worker proxy keeping tokens out of the browser; a mutual-availability algorithm (freeBusy interval inversion, two-pointer intersection) cut scheduling time from ~10 minutes to under 1 minute.

SKILLS

Languages: Python, Java, JavaScript (ES6+), SQL, HTML/CSS

AI/ML: PyTorch, TensorFlow, NumPy, Retrieval-Augmented Generation (RAG), Large Language Models (LLMs), Computer Vision (CNNs, OpenCV), PCA, Feature Engineering, Vector Search (pgvector, HNSW)

Backend & Cloud: Flask, Node.js, REST APIs, OAuth2, PostgreSQL, WebSockets, AWS (EC2), Google Cloud (GCP), Cloudflare Workers, Netlify

Practices: Data Structures & Algorithms, Concurrency, Continuous Integration (CI) / Testing, Dependency Injection, Git/GitHub